

NT-C-022 | Advanced Excel | Course Knowledge Profile

Identity, purpose, prerequisites, scope and boundaries

Category	Data & Office Productivity
Target learners	Learners needing advanced spreadsheet, MIS and reporting skills.
Prerequisites	Basic Excel navigation and formulas.
Approved duration	6-8 weeks / 50-80 guided hours.
Final outcome	Build accurate, auditable Excel models, reports and dashboards using advanced formulas and data tools.
Assessment profile	Data/AI

Approved tools / platforms

Microsoft Excel, Power Query and optional macro/VBA basics; current Microsoft 365 features where available.

Knowledge boundaries

Include	Quality rule	Exclude / escalate
The eight approved modules, four sections, named tools, projects and mapped entry-level roles.	Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.	Reject data leakage, invented metrics, unsupported causal claims and hidden manual corrections.
Current product/platform procedures only when version and date are known.	Use primary documentation and record the source version.	Do not invent current commands, prices, tax rates, policies or certification status.
Evidence-based career guidance.	Cite tests, projects, capstone, viva and verified resume evidence.	No guaranteed job, placement, salary, interview or employer claim.

LLM answer pattern: identify the course and version, answer from approved records, cite record IDs, distinguish verified facts from guidance, and state any missing evidence.

NT-C-022 | Advanced Excel | NT-C-022-S01 Deep Knowledge

Modules 1-2 | objectives, concepts, evidence and remediation

Section competency	References, logic, text, date and error handling; XLOOKUP, INDEX/MATCH and dynamic arrays
Completion evidence	Formula exercises; Lookup workbook
Section weight	20% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 1 - Formula foundation

Objective: References, logic, text, date and error handling

Observable evidence: Formula exercises

Concept	Approved knowledge statement	Application behaviour	Evidence
References	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when references is appropriate; apply it in a Advanced Excel task; verify the result.	Formula exercises
logic	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when logic is appropriate; apply it in a Advanced Excel task; verify the result.	Formula exercises
text	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when text is appropriate; apply it in a Advanced Excel task; verify the result.	Formula exercises
date	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when date is appropriate; apply it in a Advanced Excel task; verify the result.	Formula exercises
error handling	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when error handling is appropriate; apply it in a Advanced Excel task; verify the result.	Formula exercises

Module 2 - Lookup methods

Objective: XLOOKUP, INDEX/MATCH and dynamic arrays

Observable evidence: Lookup workbook

Concept	Approved knowledge statement	Application behaviour	Evidence
XLOOKUP	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when xlookup is appropriate; apply it in a Advanced Excel task; verify the result.	Lookup workbook
INDEX/MATCH	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when index/match is appropriate; apply it in a Advanced Excel task; verify the result.	Lookup workbook
dynamic arrays	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when dynamic arrays is appropriate; apply it in a Advanced Excel task; verify the result.	Lookup workbook

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Formula exercises; Lookup workbook, return to Modules 1-2, complete one guided practice and then use a new equivalent test form.

NT-C-022 | Advanced Excel | NT-C-022-S02 Deep Knowledge

Modules 3-4 | objectives, concepts, evidence and remediation

Section competency	Structured tables, validation and protection; Text-to-columns, remove duplicates and quality checks
Completion evidence	Controlled data sheet; Clean dataset
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 3 - Tables and controls

Objective: Structured tables, validation and protection

Observable evidence: Controlled data sheet

Concept	Approved knowledge statement	Application behaviour	Evidence
Structured tables	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when structured tables is appropriate; apply it in a Advanced Excel task; verify the result.	Controlled data sheet
validation	Checking input or output against defined rules before trusting it. In Advanced Excel, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when validation is appropriate; apply it in a Advanced Excel task; verify the result.	Controlled data sheet
protection	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when protection is appropriate; apply it in a Advanced Excel task; verify the result.	Controlled data sheet

Module 4 - Data cleaning

Objective: Text-to-columns, remove duplicates and quality checks

Observable evidence: Clean dataset

Concept	Approved knowledge statement	Application behaviour	Evidence
Text-to-columns	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when text-to-columns is appropriate; apply it in a Advanced Excel task; verify the result.	Clean dataset
remove duplicates	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when remove duplicates is appropriate; apply it in a Advanced Excel task; verify the result.	Clean dataset
quality checks	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when quality checks is appropriate; apply it in a Advanced Excel task; verify the result.	Clean dataset

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Controlled data sheet; Clean dataset, return to Modules 3-4, complete one guided practice and then use a new equivalent test form.

NT-C-022 | Advanced Excel | NT-C-022-S03 Deep Knowledge

Modules 5-6 | objectives, concepts, evidence and remediation

Section competency	PivotTables, PivotCharts, slicers and grouping; KPIs, charts and layout
Completion evidence	Interactive analysis; Management dashboard
Section weight	25% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 5 - Pivot analysis

Objective: PivotTables, PivotCharts, slicers and grouping

Observable evidence: Interactive analysis

Concept	Approved knowledge statement	Application behaviour	Evidence
PivotTables	An interactive aggregation used to summarise and explore tabular data. In Advanced Excel, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when pivottables is appropriate; apply it in a Advanced Excel task; verify the result.	Interactive analysis
PivotCharts	An interactive aggregation used to summarise and explore tabular data. In Advanced Excel, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when pivotcharts is appropriate; apply it in a Advanced Excel task; verify the result.	Interactive analysis
slicers	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when slicers is appropriate; apply it in a Advanced Excel task; verify the result.	Interactive analysis
grouping	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when grouping is appropriate; apply it in a Advanced Excel task; verify the result.	Interactive analysis

Module 6 - Dashboard design

Objective: KPIs, charts and layout

Observable evidence: Management dashboard

Concept	Approved knowledge statement	Application behaviour	Evidence
KPIs	A defined metric tied to a business objective and decision. In Advanced Excel, it must be demonstrated through an observable task, not only recalled as a definition.	Explain when kpis is appropriate; apply it in a Advanced Excel task; verify the result.	Management dashboard
charts	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when charts is appropriate; apply it in a Advanced Excel task; verify the result.	Management dashboard
layout	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when layout is appropriate; apply it in a Advanced Excel task; verify the result.	Management dashboard

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Interactive analysis; Management dashboard, return to Modules 5-6, complete one guided practice and then use a new equivalent test form.

NT-C-022 | Advanced Excel | NT-C-022-S04 Deep Knowledge

Modules 7-8 | objectives, concepts, evidence and remediation

Section competency	Import, transform, combine and refresh; Macros basics, audit and documentation
Completion evidence	Repeatable query; Automated reporting workbook
Section weight	30% of the weighted mock-test average
Assessment	10 concept MCQ (30 marks) + 4 interpretation scenarios (30) + 1 data/model practical (40)

Module 7 - Power Query

Objective: Import, transform, combine and refresh

Observable evidence: Repeatable query

Concept	Approved knowledge statement	Application behaviour	Evidence
Import	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when import is appropriate; apply it in a Advanced Excel task; verify the result.	Repeatable query
transform	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when transform is appropriate; apply it in a Advanced Excel task; verify the result.	Repeatable query
combine	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when combine is appropriate; apply it in a Advanced Excel task; verify the result.	Repeatable query
refresh	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when refresh is appropriate; apply it in a Advanced Excel task; verify the result.	Repeatable query

Module 8 - Automation capstone

Objective: Macros basics, audit and documentation

Observable evidence: Automated reporting workbook

Concept	Approved knowledge statement	Application behaviour	Evidence
Macros basics	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when macros basics is appropriate; apply it in a Advanced Excel task; verify the result.	Automated reporting workbook
audit	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when audit is appropriate; apply it in a Advanced Excel task; verify the result.	Automated reporting workbook
documentation	A defined competency within Advanced Excel. The learner must explain its purpose, apply it in an appropriate scenario, verify the result and produce the specified evidence.	Explain when documentation is appropriate; apply it in a Advanced Excel task; verify the result.	Automated reporting workbook

Remediation: If the learner cannot explain the purpose, complete the stated task or provide Repeatable query; Automated reporting workbook, return to Modules 7-8, complete one guided practice and then use a new equivalent test form.

NT-C-022 | Advanced Excel | Skill Ontology & Dependency Map

Atomic skills, learning order and evidence strength

Skill ID	Competency	Depends on	Evidence	Type
NT-C-022-SK01	Formula foundation	Prerequisite foundation	Formula exercises	Critical
NT-C-022-SK02	Lookup methods	M01 plus earlier foundations	Lookup workbook	Critical
NT-C-022-SK03	Tables and controls	M02 plus earlier foundations	Controlled data sheet	Supporting
NT-C-022-SK04	Data cleaning	M03 plus earlier foundations	Clean dataset	Supporting
NT-C-022-SK05	Pivot analysis	M04 plus earlier foundations	Interactive analysis	Supporting
NT-C-022-SK06	Dashboard design	M05 plus earlier foundations	Management dashboard	Critical
NT-C-022-SK07	Power Query	M06 plus earlier foundations	Repeatable query	Supporting
NT-C-022-SK08	Automation capstone	M07 plus earlier foundations	Automated reporting workbook	Critical

Evidence-strength hierarchy

Strength	Evidence source	Use
High	Trainer-observed practical, viva, working source/project files and reproducible output.	May support verified mastery when rubric threshold is met.
Medium	Scored mock tests, reviewed assignments and documented portfolio artifacts.	Supports mastery with corroboration.
Low	Resume claim, self-report, attendance or video completion without demonstration.	May trigger verification; cannot alone support Apply Now.

CareerTours must not treat attendance, time spent or content opened as skill mastery.

NT-C-022 | Advanced Excel | Assessment Knowledge & Sample Items

Non-live examples for LLM training and evaluation

Sample ID	Type	Question	Approved answer / rubric anchor
NT-C-022-S01-EX01	Evidence/application	What evidence best demonstrates the combined competency of Formula foundation and Lookup methods?	The learner should provide Formula exercises; Lookup workbook and explain how the work applies References, logic, text, date and error handling; XLOOKUP, INDEX/MATCH and dynamic arrays. Recall alone is insufficient.
NT-C-022-S01-EX02	Scenario/remediation	A learner reports completing Lookup methods but cannot produce Lookup workbook. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-022-S01, request Lookup workbook, and allow a new equivalent test only after remediation.
NT-C-022-S02-EX01	Evidence/application	What evidence best demonstrates the combined competency of Tables and controls and Data cleaning?	The learner should provide Controlled data sheet; Clean dataset and explain how the work applies Structured tables, validation and protection; Text-to-columns, remove duplicates and quality checks. Recall alone is insufficient.
NT-C-022-S02-EX02	Scenario/remediation	A learner reports completing Data cleaning but cannot produce Clean dataset. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-022-S02, request Clean dataset, and allow a new equivalent test only after remediation.
NT-C-022-S03-EX01	Evidence/application	What evidence best demonstrates the combined competency of Pivot analysis and Dashboard design?	The learner should provide Interactive analysis; Management dashboard and explain how the work applies PivotTables, PivotCharts, slicers and grouping; KPIs, charts and layout. Recall alone is insufficient.
NT-C-022-S03-EX02	Scenario/remediation	A learner reports completing Dashboard design but cannot produce Management dashboard. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-022-S03, request Management dashboard, and allow a new equivalent test only after remediation.
NT-C-022-S04-EX01	Evidence/application	What evidence best demonstrates the combined competency of Power Query and Automation capstone?	The learner should provide Repeatable query; Automated reporting workbook and explain how the work applies Import, transform, combine and refresh; Macros basics, audit and documentation. Recall alone is insufficient.
NT-C-022-S04-EX02	Scenario/remediation	A learner reports completing Automation capstone but cannot produce Automated reporting workbook. What should CareerTours do?	Mark the evidence Not Yet Verified, assign practice from NT-C-022-S04, request Automated reporting workbook, and allow a new equivalent test only after remediation.

Live bank rule: Generate at least three times the live quantity for each section using the approved Data/AI blueprint. Human-review every scored item before initial production release.

NT-C-022 | Advanced Excel | Labs, Projects & Scoring Rubrics

Practical proof of learning and ownership

Level	Approved project	Skills evidenced	Required deliverables
Mini 1	Data-cleaning workbook	Validation and quality checks	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Mini 2	MIS dashboard	Pivots, charts and KPIs	Brief; working/source file or reproducible environment; output; validation notes; reflection.
Capstone	Monthly reporting pack	Power Query, formulas and controls	Brief; working/source file or reproducible environment; output; validation notes; reflection.

Standard project rubric

Dimension	Weight	What earns credit
Problem framing	15%	Clear objective, user/business need, assumptions and success criteria.
Technical/design accuracy	25%	Correct methods, appropriate tools and sound decisions.
Completeness	15%	Required features, assets, data or workflow are present.
Testing and validation	15%	Important normal, edge, quality or failure cases are checked.
Evidence and reproducibility	15%	Source/working files, setup, inputs, outputs and version information are available.
Documentation	5%	Concise structure, instructions and limitations.
Viva / ownership	10%	Learner can explain decisions, demonstrate work and respond to a change request.

Require data provenance, cleaning decisions, train/test separation, suitable metrics, reproducibility and responsible interpretation.

NT-C-022 | Advanced Excel | Career Roles, Evidence & Gap Actions

Explainable top-three occupation matching

Rank	Occupation	Default weighted skills	Minimum evidence	Gap action
1	MIS Executive	Excel 34% [critical]; pivots 33% [critical]; reporting 33%	Management workbook	If a critical skill is below 60 or management workbook is missing, return Improve First or Needs Evidence with the linked module/project.
2	Reporting Executive	Data quality 34% [critical]; formulas 33% [critical]; dashboards 33%	Repeatable report	If a critical skill is below 60 or repeatable report is missing, return Improve First or Needs Evidence with the linked module/project.
3	Junior Data Analyst - Excel	Cleaning 34% [critical]; lookup 33% [critical]; analysis 33%	Documented analysis	If a critical skill is below 60 or documented analysis is missing, return Improve First or Needs Evidence with the linked module/project.

Course readiness	45% section mocks + 20% mini projects + 25% capstone + 10% viva.
Role readiness	45% role-relevant mastery + 25% projects/capstone + 20% aligned mocks + 10% verified resume evidence.
Evidence confidence	Show separately; Apply Now requires at least 70.
Resume extraction	Use only relevant skills, projects, education, experience and portfolio evidence; minimise personal data before external model processing.

A student may be Apply Now for one mapped occupation and Improve First for another. Return independent evidence and gaps for each role.

NT-C-022 | Advanced Excel | FAQs, Misconceptions & LLM Response Pack

Course-specific retrieval and response patterns

Approved FAQ	Answer
Who is Advanced Excel for?	Learners needing advanced spreadsheet, MIS and reporting skills.
What should I know before starting?	Basic Excel navigation and formulas.
How long is the approved learning journey?	6-8 weeks / 50-80 guided hours.
What will I be able to demonstrate?	Build accurate, auditable Excel models, reports and dashboards using advanced formulas and data tools.
Does completion guarantee a job?	No. Completion and role readiness are evidence-based guidance; employment, placement, salary and employer selection are not guaranteed.
What if I fail a section?	CareerTours identifies the failed skill tags and module, assigns targeted practice, and permits an equivalent retest after remediation.

Misconception	Approved correction
Attendance proves mastery	False - require tests and observable practical/project evidence.
One total score proves every role fit	False - calculate each occupation using its weighted skills and evidence.
The LLM can add missing details	False - retrieve approved records or state that evidence is insufficient.
Course completion guarantees employment	False - never claim guaranteed job, placement, salary or interview.
Advanced Excel knowledge never changes	False - version-sensitive tools, laws, taxes, security and platform procedures require review.

Retrieval metadata

```
course_code=NT-C-022; corpus_version=3.0;
record_types=course_profile,module,section,skill,assessment_sample,project,occupation_mapping,faq,misconception;
approval_status=approved; effective_date=2026-08-12
```